LAB # 02

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Section: BSCS(C) 1st

Lab Objective:

Learning datatypes, variables, contants, arithemtic operators.

TASK - 1:

(A)

```
#include <iostream>
using namespace std;
int main()
{
    char letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    return 0;
}
```

output

```
A
B
```

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(B)

```
#include<iostream>
using namespace std;
int main ()
{
    char letter;
    letter = 65;
    cout << letter << '\n';
    letter 66;
    cout << letter << '\n';
    cin >> letter;
    return 0;
}
```

Output

A
A (65=A, 66=B)

(C)

```
#include<iostream>
using namespace std;
int main ()
{
    int letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin >> Letter;
    return 0;
}
```

Output

65
A

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Task 3:-

```
#include<iostream>
using namespace std;
int main ( )
        {
          int val1, val2, val3;
          cout << "Enter val 1:";
          cin >> val1;
         cout<< Enter val2 :";
          cin >> val2;
          cout << "Enter val 3:";
          cin >> val3;
          cout << "\In order:" << val1 << val2 << val3 << endl;
          cout << "In reverse:" << val3 << val2 << val1 << endl;
           return 0;
           }
```

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Task-4: Distance between two points

Formula : $d = \sqrt{(x2-x1)^2+(y2-y1)^2}$

```
#include <iostream>
#include <cmath>
using namespace std;
int main ()
{
double x1,y1,x2,y2;
cout<<"Enter x1 y1 x2 y2":;
cin >> x1>>y1>> x2>>y2;
double d = Sqrt(power(x2-x1,2)+pow(y2-y1,2));
cout<< "Distance" = << d <<end 1;
return 0;
}
```

Task-5 Area programs

1) Circle

```
#include <iostream>
#using namespace std,
int main ()
{
double r, area,
const double PI = 3.14159;
cout << "Enter radius";
cint >> r;
area = PI * r * r;
cout << "Area of Circle: "<< area,
```

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2) Rectangle:

```
#include <iostream>
using namespace std;
int main()
{
    double length, width;
    cout<< "Enter length and width:";
    cin >> length>> width;
    cout<< "Area of Rectangle="<< length* width;
    return 0;
}
```

---

3) Triangle

```
#include <iostream>
using namespace std;
int main()
{
    double base, hight;
    cout<< "Enter base and hight:";
    cin >> base >> height;
    cout<< "Area of Triangle="<< base* hight*0
    return 0;
}
```

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Date = 07/10/26

Home Task

program 01

```
#include <iostream>
using namespace std;
int main ()
{
    int mark = 50;
    if (mark >= 50)
    cout << "pass";
    else if (mark < 50)
    cout << "fail",
    return 0;
}
```

---

program 02

```
#include<iostream>
using namespace std;
int main ()
{
    int mark=50;
    if (mark <0):
    cout "positive";
    else if (mark >0)
    cout " Negative";
    return0;
}
```

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program 03

```
int mark = 50;
if (mark < 0)
cout << "odd";
else if (mark > 50)
cout << "even";
return 0;
}
```

---

program 04

```
#include <iostream>
using namespace std;
int main()
{
int mark = 65;
if (mark < = 0)
cout << "grater";
else if (mark >= 0)
cout << "greater then equal";
return 0;
}
```

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